

Csenge Petak

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ORCID

Research Statement

I am a theoretical evolutionary biologist with a deep-rooted interest in fundamental laws governing biology. I have worked on computational models of gene regulatory networks to investigate their dynamics and evolution, and their influence on evolvability in variable environments. I am interested in continuing research in this general direction combining complex systems research with theoretical biology to investigate the emergence of core biological behaviours, pattern formation and evolution.

Education

- **PhD in Biology**
University of Vermont, Burlington, VT, USA 2019 - 2025
Studied adaptation to variable environments using sea urchins and simulations. Built and analysed computational models of gene regulatory networks and their evolution, fitness landscapes, and target spatial-temporal developmental patterning generated by cellular automata rules.
Advisor: Dr. Melissa Pespeni
- **Complex Systems and Data Science Graduate Certificate**
Vermont Complex Systems Institute, Burlington, VT, USA 2020 - 2022
Training in computational and mathematical techniques to describe and understand complex natural and sociotechnical systems. Classes taken: Principles of Complex Systems, Modelling Complex Systems, Evolutionary Computation, Data Science I, Data Lab. GPA: 4.0
- **BSc in Biology**
University of Southampton, Southampton, UK 2016 - 2019
Undergraduate degree with special focus on evolutionary, developmental, and molecular biology. First-Class Honours and Dean's List Award.
- **Extra courses**.....
 - **PH526x: Using Python for Research**
HarvardX certificate 2018
 - **CS50's Introduction to Computer Science**
HarvardX certificate 2016 - 2017

Research experience

- **Postdoctoral Researcher**
Friedrich Miescher Laboratory of the Max Planck Society, Tuebingen, Germany 2026 -
Beyond averages: the evolution of transcriptional variability and its role in adaptation and evolutionary innovation. Part of the Genomic Basis of Evolutionary Innovations (GEvol) DFG Priority Programme.
Advisor: Dr. Luisa Pallares

- **Research Assistant - Phyllotaxis Computational Modelling**
University of Cambridge, Cambridge, UK 2023
 Investigated different models of stripe formation and the ways in which plant morphogens influence development when evolved to produce different patterns. Lead by Dr. Renske Vroomans at the Sainsbury Laboratory Cambridge University (SLCU).
- **Research Collaborator - Evolution of Biological Noise**
Brown University, Providence, RI, USA 2020 - 2024
 Theoretical project investigating evolvability, variability and the population genetics of alleles that influence biological noise with collaborators Dr. Daniel M. Weinreich and Dr. Ignacio G. Bravo.
- **Research Trainee - Biological Data Science**
University of Vermont, Burlington, VT, USA 2019 - 2025
 Graduate Training Program - formerly known as the Quantitative Evolutionary STEM Training program, funded by the United States National Science Foundation.
- **Wellcome Trust Biomedical Vacation Scholarship**
University of Southampton, Southampton, UK 2018
 Applied and received funds to examine the role of alginate in the attachment of *Pseudomonas aeruginosa* on urinary catheters by comparing biofilm formation-defective mutant strains.
- **Undergraduate Research Assistant**
University of Southampton, Southampton, UK 2017
 Assisted with research on speciation through homoploid hybridisation in *Argyranthemum* in an Evolutionary Genomics and Transcriptomics lab.

Additional notable projects.....

- **Modularity in complex contagion models**
Python simulation 2019
 Investigated the emergence and function of modularity in grown networks with threshold-based activation functions. Collaborators: Dr. Lapo Frati and Karl Dejong Kaiser.
- **Computational model of mutation rate evolution**
Python simulation 2018
 Explored the indirect evolution of mutation rates in a population of food-seeking agents moving on a toroidal lattice. Collaborators: Dr. Lapo Frati and Dr. Richard A. Watson.

Publications and preprints

- * **Petak, C.**, Sadler, D. E., Nunez, J. C., and Pespeni, M. H. (2026). Evidence of Adaptation in Structural Variants among Wild Populations of the purple sea urchin, *Strongylocentrotus purpuratus*. [bioRxiv](#). In review at Genome Biology and Evolution.
- Weinreich, D. M., Sgouros, T., Raynes, Y., Burtsev, H., Chang, E., Rajakumar, S., ... and **Petak, C.** (2026). Wanted: A population genetic theory of biological noise regulation. *PLoS genetics*, 22(3), e1012066.
- * **Petak, C.**, Frati, L., Vroomans, R. M., Pespeni, M. H., and Cheney, N. (2025). The variability of evolvability: properties of dynamic fitness landscapes determine how phenotypic variability evolves. [Proceedings of the National Academy of Sciences \(PNAS\)](#)
- Frati, L., **Petak, C.**, and Cheney, N. (2024). Networks of Binary Necklaces Induced by Elementary Cellular Automata Rules. *Proceedings of the 2024 Artificial Life Conference*. (pp. 117). ASME.
- * **Petak, C.**, Frati, L., Brennan, R. S., and Pespeni, M. H. (2023). Whole-Genome Se-

quencing Reveals That Regulatory and Low Pleiotropy Variants Underlie Local Adaptation to Environmental Variability in Purple Sea Urchins. *The American Naturalist*, 202(4), 571–586.

- * **Peták, C.**, Frati, L., Pespeni, M. H., and Cheney, N. (2023). Coping with seasons: evolutionary dynamics of gene networks in a changing environment. *Proceedings of the Companion Conference on Genetic and Evolutionary Computation*, 163–166.
- Bunford, N., Csibra, B., **Peták, C.**, Ferdinandy, B., Miklósi, Á., and Gácsi, M. (2019). Associations among behavioral inhibition and owner-rated attention, hyperactivity/impulsivity, and personality in the domestic dog (*Canis familiaris*). *Journal of comparative psychology*, 133(2), 233.

Awards

Outstanding Doctoral Dissertation of the Year Award	1 st place	2026
University of Vermont Computer Science Fair	3 rd place	2023
Graduate Student Senate Most Innovative Research Award	1 st place	2021
University of Southampton Faculty Merit Award	1 st place	2018
Graduate College Conference and Chair's Award	\$900	2024
Quantitative Evolutionary STEM Training program	\$10,500	2023
Dr. Roberto Fabri Fialho Research Award	\$1,850	2022
John Wheeler Graduate Student Research and Development Award	\$1,600	2021
Wellcome Trust Biomedical Vacation Scholarship	£2,000	2018

Technical skills

- **Programming languages:** Python, R, C++, $\text{\LaTeX}$, Bash
- **Computational modelling:** Building computational models of complex systems, systems of ordinary differential equations, agent-based computing, network dynamics, cellular automata, genetic algorithms, training and using artificial neural networks.
- **Computational tools:** Data science techniques for statistics and visualization of high dimensional data and networks. Scheduling and running jobs in parallel on high-performance computing clusters, using virtual environments for package management, making basic websites with Quarto, and version control with git.
- **Wet lab skills:** DNA extraction, primer design, PCR amplification and purification of product, gel electrophoresis, molecular cloning, heat-shock transformation, inoculation of agar plates and broths, fluorescent staining, using epifluorescence confocal microscope, microinjection of live egg cells, maintenance and experimental handling of live research animals.

Conferences and talks

- **Artificial Life, Intelligence, Complexity & Evolution (ALICE) workshop**
Presenter 2026
 Title: The Challenges of Life: Training neural networks to learn different life-like 2D cellular automata rules
- **Artificial Life Conference (Alife)**
Attended, Co-Author 2024, 2025
 Title: Networks of Binary Necklaces Induced by Elementary Cellular Automata Rules.
- **The Allied Genetics Conference (TAGC)**
Presenter 2024

Title: Non-heritable yet evolvable: increased developmental noise can be selected for despite average negative effect

- **The Genetic and Evolutionary Computation Conference (GECCO)**
Presenter 2023
Title: Coping with seasons: evolutionary dynamics of gene networks in a changing environment
- **Developmental Biology of Sea Urchins and other Marine Invertebrates XXVI**
Presenter 2022
Title: Whole-genome sequencing shows the role of gene regulation in local adaptation to environmental variability in purple sea urchins
- **SSE, SSB, ASN joint Evolution Conference**
Presenter 2021
Title: Local adaptation to environmental variability through the evolution of gene regulation in a heterogeneous seascape
- **University of Vermont Student Research Conference**
Presenter 2021 - 2024
- **Evolutionary Systems Biology**
Attended 2020, 2022
- **University of Southampton Natural History Society Symposium**
Invited speaker 2018

Teaching experience

- **Graduate Teaching Assistant**
Vermont Complex Systems Institute, Burlington, VT, USA 2022 - 2025
Graduate courses: Data Science I, Evolutionary Computation, Modelling Complex Systems
Undergraduate courses: Introduction to Programming
- **Graduate Teaching Assistant**
University of Vermont Biology Department, Burlington, VT, USA 2019 - 2023
Graduate courses: Ecological Genomics
Undergraduate courses: Genetics, Ecology and Evolution, Comparative Physiology

Service

- **Reviewer** for the Conference on Artificial Life and the American Naturalist.
- **Mentor** for undergraduate researchers Mackenzie E. Kerner, Julia Footh, Kolding Rasmussen, Meg Hall and Cora Deininger.
- **Program support advisor** for the Thermofly 2022 Summer Undergraduate Research Experience
- **Member** of the QuEST Evolutionary Genomics Interest Group Network
- **Demonstrator** at the Southampton Science and Engineering Day - evolutionary biology/fossils booth
- **Member** of the Darwin Journal Club at the University of Southampton
- **Current and Past Memberships** of the following societies: Complex Systems Society (CSS), Society for the Study of Evolution (SSE), Society for Molecular Biology and Evolution (SMBE), EchinoClub, Genetics Society of America

Extra-curricular activities

Rock climbing, hiking, geocaching

References

Name: Dr. Melissa Pespeni
Institution: University of Vermont
Email Address: Melissa.Pespeni@uvm.edu
Reference Description: PhD advisor 2019 - 2025

Name: Dr. Daniel Weinreich
Institution: Brown University
Email Address: Daniel_Weinreich@brown.edu
Reference Description: PhD committee member and collaborator since 2019

Name: Dr. Nick Cheney
Institution: University of Vermont
Email Address: ncheney@uvm.edu
Reference Description: PhD committee member and collaborator since 2021